Code Layout, Readability and Reusability

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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Repository layout

Repository: https://github.com/thakurkumkum/travel-itinerary-planner

Path / area	Responsibility
frontend/src/App (App.tsx / App.jsx)	Application-level page and state coordination; holds destination state across views.
frontend/src/pages or components	Reusable UI: Home, Planner, Itinerary and My Trips views.
frontend/src/services or API client	Centralised calls to the FastAPI backend and the Supabase client.
frontend/package.json	Frontend scripts and dependencies.
backend/app/main.py	FastAPI application setup, CORS middleware and route registration.
backend/app/api/routes/itineraries.py	Itinerary generation endpoint.
backend/app/schemas/itinerary.py	Pydantic request / response models.
backend/app/config.py	Backend configuration and environment handling.
backend/app/supabase_client.py	Supabase client module.
backend/requirements.txt	Python dependencies installed on Render.
backend/.python-version	Pins Python 3.11.9 for the hosted runtime.
.env.example	Names and placeholder values of required variables; never real secrets.

Maintainability principles

- Keep page components focused and share common UI components.
- Centralise the API base URL (VITE_API_BASE_URL); avoid duplicated endpoint strings.
- Validate input on both client and server using typed Pydantic schemas.
- Show user-friendly errors; never hard-code secrets.

Readability and reuse checks

Check	Practice followed
Naming	Descriptive names for components, handlers, schemas and variables.
Separation of concerns	UI, API calls, business logic and persistence are separated.
Error handling	HTTP errors, network failures and invalid payloads are handled with readable messages.
Configuration	Environment-specific settings come from environment variables.
Documentation	README describes setup, environment variables, run commands and deployment.
Tests	Manual production smoke tests recorded; automated unit / integration tests are planned future work.

Exact filenames follow the current GitHub branch; the paths above are the logical layout of the project.